

Concours Commun Mines-Ponts (CCMP) 2023

Physics I Exam, PC Track

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Reviewed and Corrected Solution

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Notes on this revision

Every numerical result of the previous version was recomputed independently and every figure was re-read. The physics and the algebra were correct throughout; the changes below are one arithmetic correction, two arguments that were incomplete, and a number of places where the figures allow a sharper answer than was given.

- **Q8, corrected.** $H = RT_0/(Mg_0) = 8.48$ km, which is 8 km to one significant figure, not 9 km. The previous value came from rounding twice ($8.48 \rightarrow 8.5 \rightarrow 9$). This does not affect $\xi = \ell/H \simeq 6$ or $\chi \simeq 1.1 \times 10^3$ km.
- **Q19, completed.** The previous argument identified the decay constant of $N(x)$ rather than computing a mean. The free-path probability density is now derived and $\langle x \rangle$ integrated.
- **Q3, quantified.** The colour bars of Figure 4 give 836 LSB (visible) and 62 LSB (filtered). The visible reading is exactly one conventional flash and the filtered reading is about twice what that flash should give, so the argument is a factor $\simeq 2$ excess, not a “much larger” ratio. The reason the 761 nm band discriminates by altitude is now stated.
- **Q1, sharpened.** The tallest peak of the emission spectrum (760 nm) is infrared and cannot explain a colour; the red appearance comes from the 640 nm to 690 nm group being the only significant emission inside the visible window.
- **Q24, disambiguated.** Figure 7 is calibrated for $\ell_{pm} = 2$ cm, not for the 1 cm quoted in the statement; the back-calculation is now shown, together with the $\mathcal{N} \simeq 32$ that a literal 1 cm would give.
- **Q7, Q9, Q11, Q12, Q13, Q15, Q17, Q22** were correct and have been strengthened: ground-track versus orbital speed, the size of the ε expansion error, numerical temperatures at 20 and 50 km, oxygen partial pressures, a self-consistent bound on v/c , a proof that $\rho = 0$ rather than an assertion, the ambient-density contrast behind the radar echoes, and the $v \simeq 2c$ reductio for the relativistic electrons.
- **Course-question rigour (Q2, Q4, Q5, Q13, Q14, Q23).** Several answers stated a standard result instead of establishing it. Gauss’ theorem is now actually applied to a spherical surface rather than invoked for its conclusion; the constancy of the orbital speed is proved from $\mathbf{v} \cdot \mathbf{g} = 0$, as the statement’s “after showing that it is constant” requires; the magnetic Lorentz term is bounded using $\mathbf{B} = (k/\omega)\mathbf{e}_z \times \mathbf{E}$ and v/v_ϕ rather than the inapplicable $B = E/c$, since the plasma is dispersive; the ion current is suppressed by an explicit mass ratio; and the doubling convention of Question 23 is stated so that Question 24 can use it. Question 2 now carries the horizon diagram the statement explicitly asks for.
- **Structure.** Section titles now follow the three parts of the statement and are in the language of the document; the notation n_{mo} matches the statement; the constants used are listed, since the statement’s data annex is missing from the available scan.

The distribution of effort follows the examiners’ own report on this paper (CCMP, *Rapport sur les épreuves écrites 2023*, §2.5). It notes that questions 4–6 and 13–16 are close to the course, so full standard derivations are expected there, while questions 1–3, 17 and 24 call for initiative and document analysis, where “constructed, precise and concise” reasoning is wanted. It also records that numerical-application questions are marked all or nothing, with the significant figures required by the statement, which is why the Question 8 rounding above is not a cosmetic point.

Checked answer key

Question	Main checked result
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- 1 The dominant *visible* emission is the 600 nm to 700 nm red group; the tallest peak of all, near 760 nm, and the 320 nm to 380 nm group both lie outside the eye’s range. From the ground the UV group is completely absorbed and the narrow O₂ band at 762 nm removes the strongest line.
- 2 Pic du Midi to the French Alps is of order 500 km; using a somewhat larger Mont Blanc distance, around 600 km, would give a horizon height closer to 30 km. In either case, sprites at 40 km to 80 km can clear the horizon, while Mont Blanc cannot in the no-refraction model.
- 3 The visible image tops out at 836 LSB, exactly one conventional flash, while the filtered image reaches 62 LSB, roughly twice the 30 LSB that flash should produce at 761 nm. The ~ 30 LSB excess must come from a second emitter high enough to escape O₂ reabsorption, i.e. a sprite.
- 4–6 At 400 km: $G_{\text{ISS}} \simeq 9 \text{ m s}^{-2}$, $v_{\text{ISS}} \simeq 8 \text{ km s}^{-1}$, $T_{\text{ISS}} \simeq 90 \text{ min}$.
- 7 A 20 km camera field is swept in about 3 s at the ground-track speed. A sprite lasting a few 0.1 s can be recorded whole, with roughly kilometre-scale spatial precision.
- 8–12 Hydrostatic model gives $H = RT_0/(Mg_0) = 8.5 \text{ km} \simeq 8 \text{ km}$ and the pressure law in the statement. The model fails mainly because the assumed linear temperature law is unrealistic above the troposphere.
- 13–16 Collisionless plasma: $\gamma_p = -ine^2/(m_e\omega)$, $\omega^2 = \omega_p^2 + c^2k^2$, and for $n = 10^{11} \text{ m}^{-3}$, $f_p \simeq 3 \text{ MHz}$.
- 17 A 2.2 MHz radar reflects where $f_p \geq 2.2 \text{ MHz}$, i.e. $n_e \gtrsim 6 \times 10^{10} \text{ m}^{-3}$. The transient radar columns are consistent with sprite-related ionization.
- 18–20 $N(x) = N_0 \exp(-n_{\text{mo}}S_{\text{eff}}x)$, $\ell_{pm} = 1/(n_{\text{mo}}S_{\text{eff}})$, and at 60 km, $\ell_{pm} \simeq 2 \text{ cm}$.
- 21–24 A resting electron gains only about 2 eV over one mean free path in a 100 V m^{-1} field. A 1 MeV electron is relativistic. The avalanche stage number is $\mathcal{N} \simeq 35$: a few tens of stages, over a scale $\mathcal{N}\ell_{pm} \simeq 0.7 \text{ m}$.
-

1 Constants used

The statement announces a formula sheet and numerical data “at the end of the problem”. That annex is missing from the available scan, so the constants used below are listed explicitly here; every numerical result in this document can be reproduced from them.

Symbol	Value	Used in
R_T	6400 km	Q2, Q4–Q6, Q8–Q10
g_0	9.8 m s^{-2}	Q4–Q6, Q8
R (gas constant)	$8.31 \text{ J mol}^{-1} \text{ K}^{-1}$	Q8
k_B	$1.38 \times 10^{-23} \text{ J K}^{-1}$	Q20
e	$1.60 \times 10^{-19} \text{ C}$	Q16, Q17, Q21
m_e	$9.11 \times 10^{-31} \text{ kg}$	Q16, Q17, Q22
ε_0	$8.85 \times 10^{-12} \text{ F m}^{-1}$	Q16, Q17
$m_e c^2$	0.511 MeV	Q22
c	$3.00 \times 10^8 \text{ m s}^{-1}$	Q13, Q15, Q22

2 Extracted figures

These are the figures used in the reasoning below. They are included so the LaTeX file can be read without switching back to the exam statement.

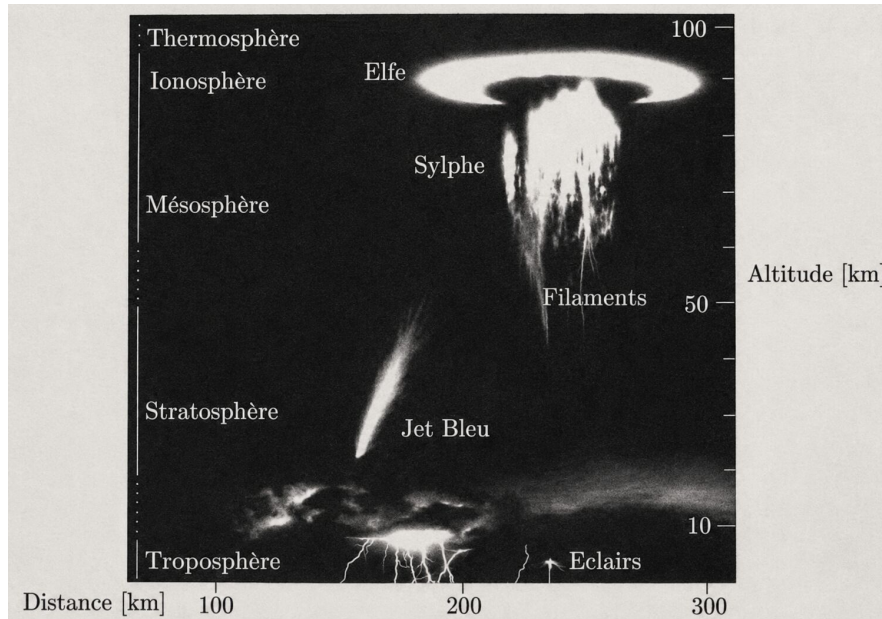


Figure 1: Figure 1: atmospheric luminous events and altitude scale.

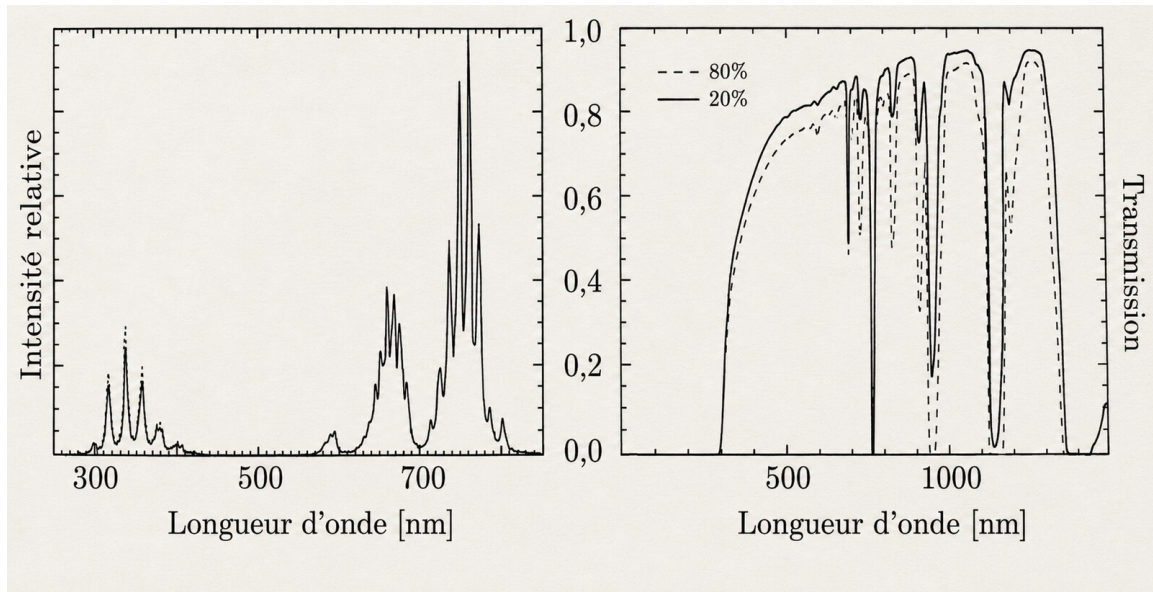


Figure 2: Figure 2: sprite emission spectrum and atmospheric transmission.

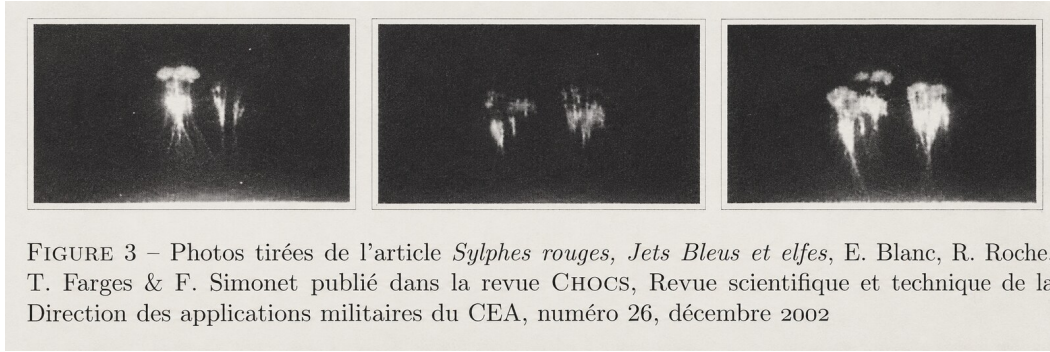


Figure 3: Figure 3: sprite photographs taken from Pic du Midi.

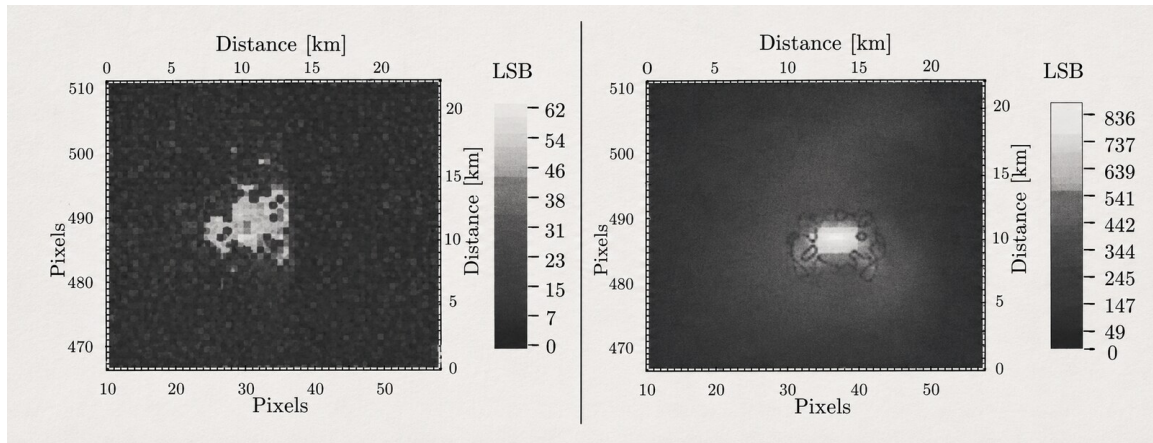


Figure 4: Figure 4: ISS filtered and visible microcamera images.

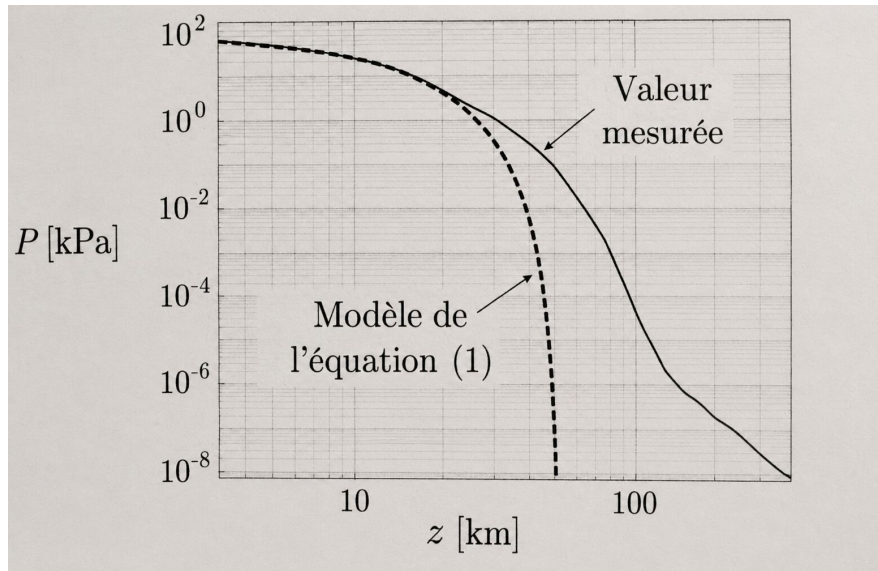


Figure 5: Figure 5: atmospheric pressure model compared with measured pressure.

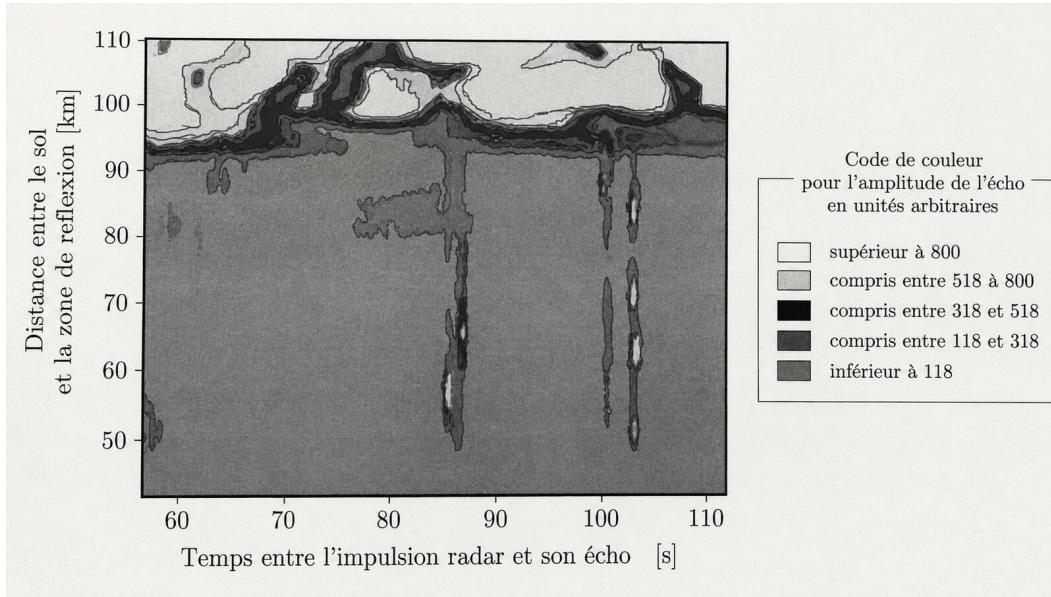


Figure 6: Figure 6: radar echo map at 2.2 MHz.

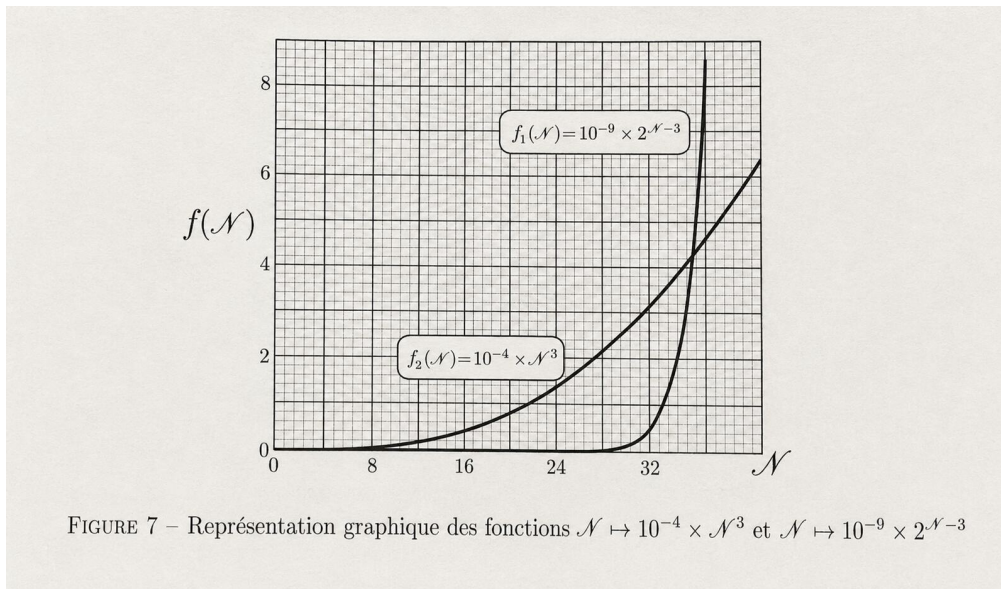
FIGURE 7 – Représentation graphique des fonctions $\mathcal{N} \mapsto 10^{-4} \times \mathcal{N}^3$ et $\mathcal{N} \mapsto 10^{-9} \times 2^{\mathcal{N}-3}$

Figure 7: Figure 7: graphical solution for the avalanche stage count.

3 Part I – Observation of sprites

Question 1 – Colour and why ground observation is difficult

The left graph in Figure 2 is an emission spectrum. It has three groups of lines: a group between about 320 nm and 380 nm, a group around 640 nm to 690 nm, and the tallest group of all around 740 nm to 780 nm.

The colour argument has to be made on the *visible* part only. The eye responds roughly between 400 nm and 700 nm, so the 320 nm to 380 nm group is ultraviolet and the tallest peak near 760 nm is near infrared: neither contributes to the perceived colour. What is left inside the visible window is the 640 nm to 690 nm group, and there is essentially no emission in the blue or green. The sprite therefore looks red.

The right graph shows the transmission of the atmosphere for light emitted at about 80 km and detected at the ground. Transmission near 1 means the light gets through; near 0 means it is absorbed. Two features matter here:

- 1) transmission collapses to zero below roughly 400 nm, so the whole ultraviolet group of the sprite spectrum never reaches the ground;
- 2) there is a very narrow but very deep notch near 762 nm, where transmission falls to zero. This is the O₂ band quoted later in the statement, and it sits exactly on the strongest emission group of the sprite.

Comparing the two dashed and solid curves also shows that the absorption bands deepen at 80 % humidity relative to 20 %, so water vapour makes matters worse.

The energy that survives is thus only the red group, already reduced by scattering and by the neighbouring bands. Added to this are two non-spectral difficulties: the events are brief, of order a few hundred milliseconds, and they occur at 40 km to 80 km above cloud tops, so a ground observer looks at them through the entire thickness of the atmosphere and usually through the thunderstorm itself.

Final answer. Inside the visible window the only significant emission is the 640 nm to 690 nm group, so sprites look red; the tallest peak (760 nm) and the 320 nm to 380 nm group are invisible to the eye. Ground observation is difficult because transmission vanishes below 400 nm and in the narrow O₂ notch at 762 nm, which lands on the strongest emission group, and because the events are brief and seen through the whole atmosphere.

Question 2 – Earth curvature and visibility from Pic du Midi

The Pic du Midi is in the Pyrenees. The French Alps and Mont Blanc are several hundred kilometres away. Among 500 km, 1000 km, and 1500 km, the correct order of magnitude for the exam is therefore

$$d \sim 500 \text{ km.}$$

A more precise distance to a specific Alpine target such as Mont Blanc can be closer to about 600 km. This does not change the conclusion; it only raises the geometric horizon estimate from about 20 km to about 30 km.

The statement asks for a diagram, so here are the assumptions it encodes: Earth is a sphere of radius R_T , the observer's own altitude is neglected as instructed, and there is no atmospheric refraction. The line of sight from the observer A is then tangent to the sphere at A . Writing B

for the Alpine ground point and C for the point of the tangent line directly above it, the height $h = BC$ is how far above the ground a target must be to clear the horizon.

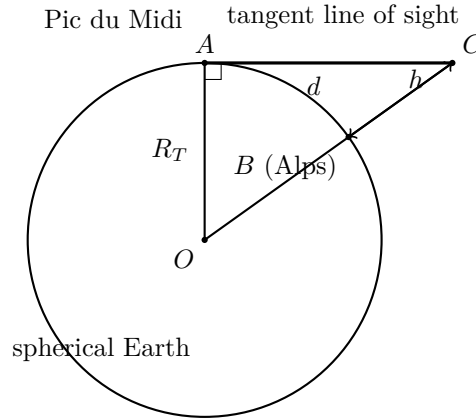


Figure 8: Tangent-horizon geometry. To the accuracy required here the map separation is identified with the tangent length AC .

The triangle OAC is right-angled at A , because AC is tangent at A . With $OA = R_T$, $OC = R_T + h$ and $AC \simeq d$, Pythagoras gives

$$(R_T + h)^2 = R_T^2 + d^2, \quad \text{so} \quad h = \sqrt{R_T^2 + d^2} - R_T = R_T \left[\sqrt{1 + \left(\frac{d}{R_T}\right)^2} - 1 \right].$$

Since $d/R_T \ll 1$,

$$h \simeq \frac{d^2}{2R_T}.$$

Reading d instead as an arc length along the surface gives $h = R_T[\sec(d/R_T) - 1]$, with the same leading term $d^2/(2R_T)$; at $d = 500$ km the two exact forms give 19.5 km and 19.6 km, so the distinction is irrelevant at one significant figure. With $d = 500$ km and $R_T = 6400$ km,

$$h \simeq \frac{(5.0 \times 10^5 \text{ m})^2}{2 \times 6.4 \times 10^6 \text{ m}} \simeq 2.0 \times 10^4 \text{ m} \simeq 20 \text{ km}.$$

Sprites lie roughly between 40 km and 80 km, so they can be above this geometric horizon. Mont Blanc is only about 4.8 km high, which is far below even the 20 km to 30 km horizon-height estimate. In the no-refraction model, Mont Blanc is not visible from Pic du Midi at this distance.

Final answer. $d \sim 500$ km for the exam order of magnitude, giving $h \sim 20$ km. A 600 km distance would give $h \sim 30$ km. Alpine sprites can be visible because they are much higher than this, but Mont Blanc cannot be seen in this simplified no-refraction geometry.

Question 3 – Why Figure 4 proves a sprite detection

Figure 4 compares two simultaneous ISS images: a narrow-band image filtered around 761 nm and a visible-light image. The statement gives a useful calibration: a normal lightning flash gives about 836 LSB in the visible camera but only about 30 LSB in the filtered camera. For ordinary lightning, the filtered/visible ratio is therefore only

$$\frac{30}{836} \simeq 3.6 \times 10^{-2}.$$

Now read the two colour bars of Figure 4. The visible image saturates at 836 LSB and the filtered image at 62 LSB. The visible reading is therefore exactly the signature of one conventional lightning flash. If that flash were the only source, the filtered camera should read about 30 LSB. It reads roughly twice that:

$$\frac{62}{836} \simeq 7.4 \times 10^{-2} \simeq 2 \times \frac{30}{836}.$$

So about 30 LSB of narrow-band signal is left over once the lightning is accounted for, and it must come from something else.

That something else has to be high. The filter is centred on 761 nm, which is an O₂ absorption band. Radiation emitted at that wavelength *below* the bulk of the atmospheric oxygen is reabsorbed on its way up, so a tropospheric flash is strongly attenuated in this band, which is precisely why its filtered signature is only 30 LSB while its visible signature is 836 LSB. A source emitting above roughly 40 km has almost no O₂ left above it and is not attenuated. The excess narrow-band signal therefore identifies an emitter at sprite altitudes.

Two further remarks support the identification. The two images are simultaneous and the structure appears at the same place in both, so it is not camera noise, which would not correlate between two independent detectors. And in the filtered image the structure is visibly more extended and more ragged than the compact saturated core of the visible image, as expected if a diffuse high-altitude emission is superposed on a point-like flash.

Final answer. The visible image reads 836 LSB, i.e. one conventional flash, but the filtered image reads 62 LSB instead of the 30 LSB that flash should give at 761 nm. The ~ 30 LSB excess in an O₂ absorption band can only come from an emitter high enough to escape reabsorption, and the two co-located simultaneous signals rule out noise. This is a sprite detection.

Question 4 – Gravity at ISS altitude

The statement asks for the assumptions to be specified. They are: Earth is spherical with a spherically symmetric mass distribution; the ISS is outside it; Earth's rotation, atmospheric drag and the fields of other bodies are neglected.

Spherical symmetry forces the field to be radial with a magnitude depending only on r ,

$$\mathbf{g}(r) = -g(r) \mathbf{e}_r,$$

so g is constant over any sphere $\mathcal{S}_r$ centred on Earth. Gauss' theorem for gravitation, the analogue of the electrostatic one with $1/\varepsilon_0 \rightarrow -4\pi G_N$, reads

$$\oint_{\mathcal{S}_r} \mathbf{g} \cdot d\mathbf{S} = -4\pi G_N M_T.$$

With $d\mathbf{S}$ outward, the left-hand side is $-g(r) 4\pi r^2$, hence

$$g(r) = \frac{G_N M_T}{r^2}.$$

This is the standard result that a spherically symmetric body acts, on the outside, like a point mass at its centre. Applying it twice, at $r = R_T$ and at $r = R_T + h_{\text{ISS}}$ with $h_{\text{ISS}} \simeq 400$ km,

$$g_0 = \frac{G_N M_T}{R_T^2}, \quad g_{\text{ISS}} = \frac{G_N M_T}{(R_T + h_{\text{ISS}})^2},$$

and dividing the second by the first:

$$G_{\text{ISS}} = g_0 \left(\frac{R_T}{R_T + h_{\text{ISS}}} \right)^2 = g_0 \left(1 + \frac{h_{\text{ISS}}}{R_T} \right)^{-2}.$$

Numerically,

$$\frac{h_{\text{ISS}}}{R_T} = \frac{400}{6400} = 0.0625,$$

so

$$G_{\text{ISS}} = \frac{9.8}{(1.0625)^2} \simeq 8.7 \text{ m s}^{-2}.$$

Final answer.

$$G_{\text{ISS}} = g_0 \left(1 + \frac{h_{\text{ISS}}}{R_T} \right)^{-2} \simeq 9 \text{ m s}^{-2}.$$

Question 5 – ISS orbital speed

The statement says “after showing that it is constant”, so this has to be proved rather than assumed. On a circular orbit the velocity is tangential while the gravitational field is radial, so the two are orthogonal and

$$\frac{d}{dt} \left(\frac{v^2}{2} \right) = \mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = \mathbf{v} \cdot \mathbf{g} = 0.$$

The speed v_{ISS} is therefore constant: gravity does no work here, it only turns the velocity.

Writing $r = R_T + h_{\text{ISS}}$, a uniform circular motion has purely centripetal acceleration, and Newton’s second law projected on $\mathbf{e}_r$ gives

$$\mathbf{a} = -\frac{v_{\text{ISS}}^2}{r} \mathbf{e}_r = \mathbf{g} = -G_{\text{ISS}} \mathbf{e}_r, \quad \text{so} \quad \frac{v_{\text{ISS}}^2}{r} = G_{\text{ISS}}.$$

Thus

$$v_{\text{ISS}} = \sqrt{G_{\text{ISS}}(R_T + h_{\text{ISS}})}.$$

Using the expression from Question 4,

$$v_{\text{ISS}} = \sqrt{\frac{g_0 R_T^2}{R_T + h_{\text{ISS}}}}.$$

Numerically,

$$v_{\text{ISS}} \simeq \sqrt{\frac{9.8(6.4 \times 10^6 \text{ m})^2}{6.8 \times 10^6 \text{ m}}} \simeq 7.7 \times 10^3 \text{ m s}^{-1}.$$

Final answer. The speed is constant because the force is radial and the velocity tangential, so $\mathbf{v} \cdot \mathbf{g} = 0$. Then

$$v_{\text{ISS}} = \sqrt{\frac{g_0 R_T^2}{R_T + h_{\text{ISS}}}} \simeq 8 \text{ km s}^{-1}.$$

Question 6 – ISS period and Kepler’s third law

One circular orbit has length $2\pi(R_T + h_{\text{ISS}})$. Therefore

$$T_{\text{ISS}} = \frac{2\pi(R_T + h_{\text{ISS}})}{v_{\text{ISS}}}.$$

Substitute the speed from Question 5:

$$T_{\text{ISS}} = 2\pi\sqrt{\frac{(R_T + h_{\text{ISS}})^3}{g_0 R_T^2}}.$$

Since $g_0 R_T^2 = GM_T$, this is also

$$T_{\text{ISS}} = 2\pi\sqrt{\frac{r^3}{GM_T}}, \quad r = R_T + h_{\text{ISS}}.$$

Squaring,

$$T_{\text{ISS}}^2 = \frac{4\pi^2}{GM_T} r^3.$$

This is Kepler’s third law: T^2 is proportional to r^3 .

Numerically,

$$T_{\text{ISS}} \simeq \frac{2\pi(6.8 \times 10^6 \text{ m})}{7.7 \times 10^3 \text{ m s}^{-1}} \simeq 5.6 \times 10^3 \text{ s} \simeq 93 \text{ min}.$$

Final answer.

$$T_{\text{ISS}} \simeq 90 \text{ min}, \quad T^2 \propto r^3.$$

Question 7 – Can the ISS cameras record a whole sprite?

The axes of Figure 4 give the field of view directly: about 22 km horizontally by 20 km vertically.

The cameras point at nadir, so the field sweeps across the emitting layer at the *ground-track* speed rather than at the orbital speed. For a target at altitude z ,

$$v_{\text{sweep}} = v_{\text{ISS}} \frac{R_T + z}{R_T + h_{\text{ISS}}} \simeq 7.7 \text{ km s}^{-1} \times \frac{6460}{6800} \simeq 7.3 \text{ km s}^{-1}$$

at $z = 60 \text{ km}$. The difference from v_{ISS} is only about 5%, so at one significant figure it changes nothing. The time to cross the field is

$$\Delta t_{\text{field}} \simeq \frac{20 \text{ km}}{7 \text{ km s}^{-1}} \simeq 3 \text{ s}.$$

A sprite lasts only a few hundred milliseconds. For example,

$$0.1 \text{ s} \Rightarrow \Delta x \simeq 0.8 \text{ km}, \quad 0.3 \text{ s} \Rightarrow \Delta x \simeq 2.4 \text{ km}.$$

These distances are much smaller than the 20 km field of view, so recording the whole event is possible if the storm lies inside the field at the right time.

For precision, the image scale is roughly

$$\frac{20 \text{ km}}{40\text{--}50 \text{ pixels}} \simeq 0.4 \text{ km/pixel to } 0.5 \text{ km/pixel}.$$

The ISS also moves by about a kilometre during a typical event. So the natural spatial precision is not metres; it is about a kilometre.

Final answer. Recording a whole sprite is possible: the field is crossed in about 2.5 s to 3 s, while the sprite lasts only a few 0.1 s. The spatial precision is roughly kilometre-scale, set both by the pixel size (~ 0.45 km) and by the ~ 1 km of ISS motion during the event. The real limitation is not the timing but the coincidence: the storm must lie inside a 20 km window during the few seconds of the pass. The temporal resolution itself cannot be quantified from the statement, since neither the frame rate nor the exposure time is given.

Part I (continued) – A variable-pressure atmosphere

Question 8 – Hydrostatic equation and scale height

The vector law of hydrostatic equilibrium is

$$\text{grad } P = \mu \mathbf{g},$$

where μ is the mass density and $\mathbf{g}$ points downward. If z is upward, this becomes

$$\frac{dP}{dz} = -\mu g(z).$$

The atmosphere is treated as an ideal gas. For molar mass M ,

$$\mu = \frac{PM}{RT(z)}.$$

The temperature model is

$$T(z) = T_0 \left(1 - \frac{z}{\ell}\right),$$

and gravity is

$$g(z) = g_0 \left(1 + \frac{z}{R_T}\right)^{-2}.$$

Therefore

$$\frac{dP}{dz} = -\frac{PM}{RT_0(1 - z/\ell)} g_0 \left(1 + \frac{z}{R_T}\right)^{-2}.$$

Define the characteristic height

$$H = \frac{RT_0}{Mg_0}.$$

Then

$$\frac{dP}{dz} = -F(z) \frac{P}{H}, \quad F(z) = \frac{1}{(1 - z/\ell)(1 + z/R_T)^2}.$$

Numerically, with $T_0 = 290$ K and $M = 29 \times 10^{-3}$ kg mol $^{-1}$,

$$H = \frac{8.31 \times 290}{0.029 \times 9.8} = 8.48 \times 10^3 \text{ m} = 8.5 \text{ km}.$$

Rounded to the single significant figure requested by the statement,

$$H \simeq 8 \text{ km}.$$

(The unrounded value 8.5 km is the one to keep for the later numerical work in Questions 9 and 10.)

Final answer.

$$\frac{dP}{dz} = -F(z) \frac{P}{H}, \quad F(z) = \frac{1}{(1 - z/\ell)(1 + z/R_T)^2}, \quad H = \frac{RT_0}{Mg_0} = 8.5 \text{ km} \simeq 8 \text{ km}.$$

Question 9 – Pressure law

The statement gives a decomposition

$$F(z) = \frac{\alpha}{1 - z/\ell} + \frac{\beta + \gamma z}{(1 + z/R_T)^2},$$

with $\varepsilon = \ell/R_T$. The exact coefficients can be written more compactly, since $1 + \varepsilon(2 + \varepsilon) = (1 + \varepsilon)^2$:

$$\alpha = \frac{1}{(1 + \varepsilon)^2}, \quad \beta = \frac{\varepsilon(2 + \varepsilon)}{(1 + \varepsilon)^2}, \quad \gamma = \frac{\alpha\varepsilon}{R_T}.$$

Here

$$\varepsilon = \frac{50}{6400} = 7.8 \times 10^{-3} \ll 1,$$

so $(1 + \varepsilon)^2 = 1 + 1.6 \times 10^{-2}$ and, to first order in ε ,

$$\alpha \simeq 1, \quad \beta \simeq 2\varepsilon, \quad \gamma \simeq \frac{\varepsilon}{R_T}.$$

The error committed is of order $2\varepsilon \simeq 1.6\%$ on α and β , far below the one-significant-figure precision requested.

The pressure equation is

$$\frac{dP}{P} = -\frac{F(z)}{H} dz.$$

Integrate from 0 to z , using $P(0) = P_0$.

First term:

$$\int_0^z \frac{du}{1 - u/\ell} = -\ell \ln \left(1 - \frac{z}{\ell} \right).$$

This gives the factor

$$\left(1 - \frac{z}{\ell} \right)^{\ell/H}.$$

For the second term, with $y = 1 + u/R_T$, one obtains

$$\int_0^z \frac{2\varepsilon + (\varepsilon/R_T)u}{(1 + u/R_T)^2} du = \ell \left[\ln \left(1 + \frac{z}{R_T} \right) + 1 - \frac{1}{1 + z/R_T} \right].$$

Here we used $\varepsilon R_T = \ell$. Also,

$$1 - \frac{1}{1 + z/R_T} = \frac{z/R_T}{1 + z/R_T}.$$

So, after multiplication by $-1/H$, the non-logarithmic part contributes

$$-\frac{\ell}{H} \frac{z/R_T}{1 + z/R_T} = -\frac{z}{(H R_T / \ell)(1 + z/R_T)}.$$

This gives both a denominator power of $1 + z/R_T$ and a small exponential correction.

Thus

$$P(z) = P_0 \left(\frac{1 - z/\ell}{1 + z/R_T} \right)^\xi \exp \left[-\frac{z}{\chi(1 + z/R_T)} \right],$$

where

$$\xi = \frac{\ell}{H}, \quad \chi = \frac{H R_T}{\ell}.$$

Numerically,

$$\xi \simeq \frac{50}{8.5} \simeq 5.9 \simeq 6, \quad \chi \simeq \frac{8.5 \times 6400}{50} \text{ km} \simeq 1.1 \times 10^3 \text{ km}.$$

Final answer.

$$P(z) = P_0 \left(\frac{1 - z/\ell}{1 + z/R_T} \right)^{\ell/H} \exp \left[-\frac{z}{(HR_T/\ell)(1 + z/R_T)} \right].$$

So $\xi = \ell/H$ and $\chi = HR_T/\ell$.

Question 10 – Useful simplification of the pressure law

For altitudes below 50 km, the ratio z/R_T is tiny:

$$\frac{50}{6400} \simeq 8 \times 10^{-3}.$$

So

$$1 + \frac{z}{R_T} \simeq 1.$$

Also $\chi \simeq 1100$ km, so

$$\exp \left[-\frac{z}{\chi(1 + z/R_T)} \right] \simeq 1$$

for $z \leq 50$ km.

Therefore the practical simplified pressure law is

$$P(z) \simeq P_0 \left(1 - \frac{z}{\ell} \right)^{\ell/H}.$$

This simplification amounts to neglecting the variation of gravity with altitude while keeping the imposed temperature law.

At very low altitude, if $z \ll \ell$, one can go further:

$$\left(1 - \frac{z}{\ell} \right)^{\ell/H} \simeq \exp \left(-\frac{z}{H} \right),$$

which is the familiar exponential barometric law.

Final answer.

$$P(z) \simeq P_0 \left(1 - \frac{z}{\ell} \right)^{\ell/H}.$$

This keeps the temperature decrease but neglects Earth-curvature / gravity variation.

Question 11 – Why the model fails near 20 km

The model was built with the temperature law

$$T(z) = T_0 \left(1 - \frac{z}{\ell} \right), \quad \ell = 50 \text{ km}.$$

This is a linear decrease at a fixed rate $T_0/\ell = 5.8 \text{ K km}^{-1}$, which is a reasonable description of the troposphere and nothing else. The real atmosphere stops cooling at the tropopause, around 10 km to 15 km, and then *warms* through the stratosphere because ozone absorbs solar ultraviolet. The numbers make the failure obvious:

$$T_{\text{model}}(20 \text{ km}) = 290 \times 0.6 = 174 \text{ K} \quad \text{against} \quad T_{\text{real}} \simeq 217 \text{ K},$$

$$T_{\text{model}}(50 \text{ km}) = 0 \text{ K} \quad \text{against} \quad T_{\text{real}} \simeq 270 \text{ K}.$$

Too low a temperature means too small a local scale height $RT(z)/(Mg)$, hence a pressure that falls too fast. This is exactly what Figure 5 shows: the dashed model curve follows the measurement up to about 20 km, then bends away below it and collapses vertically at $z = \ell = 50 \text{ km}$, where the model temperature reaches absolute zero and the factor $(1 - z/\ell)^\xi$ vanishes.

Final answer. The linear temperature law is a troposphere-only model. Above the tropopause ($\sim 10 \text{ km}$ to 15 km) the real temperature stops falling and then rises, while the model keeps cooling to 0 K at $z = \ell$. The model temperature is therefore far too low, the scale height too small, and the predicted pressure collapses well below the measured one from about 20 km upward.

Question 12 – Observation from Earth and from the ISS

The oxygen molar fraction is assumed constant:

$$x_{\text{O}_2} \simeq 0.20.$$

Therefore the oxygen partial pressure is

$$P_{\text{O}_2}(z) = x_{\text{O}_2}P(z) \simeq 0.20P(z).$$

so the oxygen profile has the same shape as the pressure profile. Two numbers are enough. At the ground,

$$P_{\text{O}_2}(0) = 0.20 \times 1 \text{ bar} = 2 \times 10^4 \text{ Pa}.$$

At 60 km the model of Question 9 must not be used: Question 11 showed it is no longer quantitatively reliable there, and in any case $z = 60 \text{ km}$ lies beyond its domain $z < \ell$. The measured value supplied by Question 20 is used instead, $P_3 = 20 \text{ Pa}$, so

$$P_{\text{O}_2}(60 \text{ km}) = 0.20 \times 20 = 4 \text{ Pa},$$

smaller by a factor 5×10^3 .

What matters for absorption is not the local partial pressure but the total amount of O_2 *along the line of sight*. A ground observer looking up at a sprite sees it through the whole oxygen column, essentially all of which lies below 20 km ; at 762 nm this column is optically thick, which is why the transmission notch in Figure 2 reaches zero.

From the ISS the geometry is reversed. The camera is at 400 km and the source is at 40 km to 80 km , so the light only has to cross the atmosphere *above* the source, where the residual pressure is a few pascals or less, i.e. thousands of times thinner than at the ground and containing a negligible O_2 column. The 761 nm band is then transparent, which is exactly why this filter can be used from orbit as a sprite discriminator (Question 3) and could not be used from the ground.

Final answer. From the ground the line of sight crosses the entire O_2 column, from 4 Pa of partial pressure at 60 km up to $2 \times 10^4 \text{ Pa}$ at the surface, and the 762 nm band is opaque. From the ISS the light only crosses the atmosphere above the sprite, where the O_2 column is negligible, so the same band becomes usable and even diagnostic.

4 Part II – An electrical atmosphere

Question 13 – Neglecting the magnetic Lorentz force

The Lorentz force on an electron is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

The relation between $\mathbf{B}$ and $\mathbf{E}$ should be taken from Faraday's law rather than assumed to be $B = E/c$: the plasma is dispersive, so its phase velocity is not c . For the plane wave $\mathbf{E} = \mathbf{E}_0 e^{i(\omega t - kz)}$,

$$\mathbf{B} = \frac{k}{\omega} \mathbf{e}_z \times \mathbf{E}, \quad \text{so} \quad B = \frac{E}{v_\varphi}, \quad v_\varphi = \frac{\omega}{k}.$$

The ratio of the two contributions to the Lorentz force is therefore

$$\frac{|q\mathbf{v} \times \mathbf{B}|}{|q\mathbf{E}|} \leq \frac{vB}{E} = \frac{v}{v_\varphi}.$$

The dispersion relation of Question 15 gives $v_\varphi \geq c$ for any propagating mode, so this bound is even tighter than v/c . The statement specifies non-relativistic electrons, $v \ll c \leq v_\varphi$, and the magnetic term is negligible.

The estimate is self-consistent with Question 14, which gives $v = eE/(m_e\omega)$ for the velocity driven by the wave:

$$\frac{v}{v_\varphi} \leq \frac{v}{c} = \frac{eE}{m_e\omega c},$$

which stays $\ll 1$ for any field amplitude and frequency of interest here: at $\omega \sim 10^7 \text{ rad s}^{-1}$ it would take $E \sim 10^9 \text{ V m}^{-1}$ to reach $v \sim c$, far above the 100 V m^{-1} scale quoted later in the statement.

Final answer. With $\mathbf{B} = (k/\omega) \mathbf{e}_z \times \mathbf{E}$, the magnetic contribution is smaller than the electric one by a factor $v/v_\varphi \leq v/c \ll 1$, so it is negligible.

Question 14 – Electron current and plasma conductivity

Under a harmonic field, a carrier of charge q and mass m reaches a velocity amplitude $|q|E/(m\omega)$, so its current density scales as $nq^2E/(m\omega)$. In a quasi-neutral, singly ionized plasma $n_i \simeq n_e$ and the charges are equal in magnitude, so the two currents differ only by the mass ratio:

$$\left| \frac{j_i}{j_e} \right| \simeq \frac{m_e}{m_i} \lesssim 10^{-4},$$

since even a proton is about 1800 times heavier than an electron, and atmospheric ions are heavier still. The ion current is therefore negligible.

Let $e > 0$ be the elementary charge; an electron has charge $-e$. The equation of motion is

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathbf{E}.$$

The field has complex time dependence $\exp(i\omega t)$, so d/dt becomes $i\omega$:

$$i\omega m_e \mathbf{v} = -e\mathbf{E}.$$

Therefore

$$\mathbf{v} = \frac{ie}{m_e\omega} \mathbf{E}.$$

The electron current density is

$$\mathbf{j} = n(-e)\mathbf{v}.$$

Thus

$$\mathbf{j} = -i \frac{ne^2}{m_e \omega} \mathbf{E}.$$

By definition, $\mathbf{j} = \gamma_p \mathbf{E}$, so

$$\gamma_p = -i \frac{ne^2}{m_e \omega}.$$

This conductivity is purely imaginary, which has a direct energetic meaning. For complex amplitudes the cycle-averaged power density delivered to the medium is

$$\langle p \rangle = \frac{1}{2} \operatorname{Re}(\mathbf{j}_0 \cdot \mathbf{E}_0^*) = \frac{1}{2} \operatorname{Re}(\gamma_p) |\mathbf{E}_0|^2 = 0,$$

because $\operatorname{Re}(\gamma_p) = 0$. The conductivity is purely reactive: with no collisions the plasma exchanges energy back and forth with the wave over a cycle but dissipates none on average. Equivalently, the velocity is in quadrature with the field, so the force does no net work.

Final answer.

$$\gamma_p = -i \frac{ne^2}{m_e \omega}.$$

It is purely imaginary, so the average absorbed power is zero in this collisionless model.

Question 15 – Wave equation and dispersion relation

Use Maxwell's equations in a plasma with current density $\mathbf{j}$:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Before combining them, note that $\nabla \cdot \mathbf{E} = 0$ here. For the wave specified in the statement this is immediate, since $\mathbf{E}_0 \perp \mathbf{e}_z$ gives

$$\nabla \cdot \mathbf{E} = -ik \mathbf{e}_z \cdot \mathbf{E} = 0.$$

It is worth checking that this is dynamically consistent rather than merely assumed. Charge conservation and Gauss' law give

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0, \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad \mathbf{j} = \gamma_p \mathbf{E},$$

so $i\omega\rho + \gamma_p\rho/\varepsilon_0 = 0$. With $\gamma_p = -ine^2/(m_e\omega)$ this reads

$$i\rho \left(\omega - \frac{\omega_p^2}{\omega} \right) = 0,$$

which forces $\rho = 0$ unless $\omega = \omega_p$. The medium therefore stays locally neutral and the wave is transverse.

Taking the curl of Faraday's law and using $\nabla \cdot \mathbf{E} = 0$ gives

$$\Delta \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial \mathbf{j}}{\partial t}.$$

Since $c^2 = 1/(\mu_0 \varepsilon_0)$,

$$\Delta \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial \mathbf{j}}{\partial t}.$$

Insert

$$\mathbf{E} = \mathbf{E}_0 e^{i(\omega t - kz)}, \quad \mathbf{j} = \gamma_p \mathbf{E}.$$

Then $\Delta \rightarrow -k^2$ and $\partial_t \rightarrow i\omega$, so

$$-k^2 + \frac{\omega^2}{c^2} = \mu_0 i\omega \gamma_p.$$

Using $\gamma_p = -ine^2/(m_e \omega)$,

$$-k^2 + \frac{\omega^2}{c^2} = \mu_0 \frac{ne^2}{m_e}.$$

Since $\mu_0 = 1/(\epsilon_0 c^2)$,

$$k^2 = \frac{\omega^2 - \omega_p^2}{c^2}, \quad \omega_p^2 = \frac{ne^2}{m_e \epsilon_0}.$$

Final answer.

$$\omega^2 = \omega_p^2 + c^2 k^2, \quad \omega_p^2 = \frac{ne^2}{m_e \epsilon_0}.$$

Question 16 – Plasma cutoff frequency

For propagation, the wave number k must be real. Since

$$k^2 = \frac{\omega^2 - \omega_p^2}{c^2},$$

we need

$$\omega > \omega_p.$$

If $\omega < \omega_p$, then $k^2 < 0$, k is imaginary, and the wave is evanescent: it cannot propagate through that plasma region.

The cutoff frequency is

$$f_p = \frac{\omega_p}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{ne^2}{m_e \epsilon_0}}.$$

For $n = 1 \times 10^{11} \text{ m}^{-3}$,

$$\omega_p = \sqrt{\frac{(10^{11})(1.60 \times 10^{-19})^2}{(9.11 \times 10^{-31})(8.85 \times 10^{-12})}} \simeq 1.8 \times 10^7 \text{ rad s}^{-1}.$$

Therefore

$$f_p = \frac{1.8 \times 10^7}{2\pi} \simeq 2.8 \times 10^6 \text{ Hz} \simeq 3 \text{ MHz}.$$

Final answer.

$$f_p \simeq 3 \text{ MHz} \quad \text{for} \quad n = 1 \times 10^{11} \text{ m}^{-3}.$$

Waves below this frequency cannot propagate through such a dense plasma.

Question 17 – Interpreting the 2.2 MHz radar map

A radar echo returns when the wave is reflected. In this plasma model, reflection happens when the local plasma frequency is at least the radar frequency.

For $f = 2.2$ MHz, the electron density needed for cutoff is

$$n_{\text{cut}} = \frac{m_e \varepsilon_0 (2\pi f)^2}{e^2}.$$

Numerically,

$$n_{\text{cut}} \simeq 6 \times 10^{10} \text{ m}^{-3}.$$

This is close to the sprite characteristic density $1 \times 10^{11} \text{ m}^{-3}$ found in Question 16.

This is the key point: the density needed to turn back a 2.2 MHz wave is of the same order as the sprite density $1 \times 10^{11} \text{ m}^{-3}$ used in Question 16, and about 10^4 times the ambient night-time density of the D region near 60 km. Any echo coming from that altitude therefore requires ionization that is not normally there.

Figure 6 shows two distinct things.

- A permanent, horizontally continuous reflecting layer above about 95 km, with the strongest amplitudes beyond 100 km. This is simply the bottom of the ionosphere and is present throughout the record.
- Narrow, quasi-vertical columns that appear only at two instants of the record and reach down to 50 km to 60 km, with local amplitude maxima (the highest amplitude class) near 55 km to 65 km and 70 km to 85 km.

Their transient character in time, their small horizontal extent, and their altitude range, which coincides with the 40 km to 80 km band where sprites occur, all point the same way: these are localized, short-lived patches of ionization created by high-altitude discharges, not permanent ionospheric structure. This is an independent, radio-frequency confirmation of the optical detections of Part I.

A remark on the figure itself: the horizontal axis is labelled as the delay between the radar pulse and its echo, in seconds, but a reflection at 50 km to 110 km corresponds to a two-way delay of only 0.3 ms to 0.7 ms. The vertical axis already carries that delay, converted into a reflection height. The horizontal axis must therefore be the elapsed observation time within the record; the label appears to be an error in the original document.

Final answer. The 2.2 MHz radar reflects only where $n_e \gtrsim 6 \times 10^{10} \text{ m}^{-3}$, comparable to the sprite density of Question 16 and far above the ambient value at these altitudes. The permanent layer above 95 km is the ionosphere; the transient columns reaching down to 50 km to 60 km are localized patches of ionization produced by sprite-related discharges.

5 Part III – Avalanche formation of a current

Question 18 – Survival equation before collision

Consider a slice between x and $x + dx$. Its volume is

$$\Sigma dx.$$

If the molecule number density is n_{mo} , the number of molecules in the slice is

$$n_{\text{mo}} \Sigma dx.$$

Each molecule presents an effective collision cross-section S_{eff} . The total effective area in the slice is therefore

$$n_{\text{mo}} \Sigma dx S_{\text{eff}}.$$

Dividing by the total section Σ , the probability that an electron collides in this slice is

$$n_{\text{mo}} S_{\text{eff}} dx.$$

If $N(x)$ electrons have reached x without collision, the number lost in the next slice is

$$dN = -N(x) n_{\text{mo}} S_{\text{eff}} dx.$$

So

$$\frac{dN}{dx} = -n_{\text{mo}} S_{\text{eff}} N.$$

Solving with $N(0) = N_0$,

$$N(x) = N_0 \exp(-n_{\text{mo}} S_{\text{eff}} x).$$

Final answer.

$$\frac{dN}{dx} = -n_{\text{mo}} S_{\text{eff}} N, \quad N(x) = N_0 \exp(-n_{\text{mo}} S_{\text{eff}} x).$$

Question 19 – Mean free path

The quantity asked for is a *mean*, so it has to be obtained from the distribution of free paths, not merely by reading off the decay constant of $N(x)$.

Write $a = n_{\text{mo}} S_{\text{eff}}$, so that $N(x) = N_0 e^{-ax}$. Among the N_0 electrons, those undergoing their first collision between x and $x + dx$ are the ones lost from N over that slice:

$$dN_{\text{coll}} = -dN = N_0 a e^{-ax} dx.$$

The probability density of the free path is therefore

$$p(x) = \frac{1}{N_0} \frac{dN_{\text{coll}}}{dx} = a e^{-ax}, \quad x \geq 0,$$

which is correctly normalised: $\int_0^\infty a e^{-ax} dx = 1$. The mean free path is then

$$\ell_{pm} = \langle x \rangle = \int_0^\infty x p(x) dx = a \int_0^\infty x e^{-ax} dx.$$

Integrating by parts,

$$\int_0^\infty x e^{-ax} dx = \left[-\frac{x}{a} e^{-ax} \right]_0^\infty + \frac{1}{a} \int_0^\infty e^{-ax} dx = 0 + \frac{1}{a^2},$$

so $\langle x \rangle = a \times a^{-2} = 1/a$, that is

$$\ell_{pm} = \frac{1}{n_{\text{mo}} S_{\text{eff}}}.$$

Equivalently, and faster, one may integrate the survival function directly: $\langle x \rangle = \int_0^\infty (N(x)/N_0) dx = \int_0^\infty e^{-ax} dx = 1/a$.

Final answer.

$$\ell_{pm} = \frac{1}{n_{\text{mo}} S_{\text{eff}}}.$$

Question 20 – Mean free path at 60 km

The value $S_{\text{eff}} \sim 1 \times 10^{-20} \text{ m}^2$ is reasonable because a molecule has a size of order $1 \times 10^{-10} \text{ m}$; an area is therefore of order

$$(10^{-10})^2 = 10^{-20} \text{ m}^2.$$

Use the ideal gas law in number-density form:

$$n_{\text{mo}} = \frac{P}{k_B T}.$$

At 60 km, the statement gives $P_3 = 20 \text{ Pa}$ and $T_3 = 260 \text{ K}$, so

$$n_{\text{mo}} = \frac{20}{(1.38 \times 10^{-23}) \times 260} \simeq 5.6 \times 10^{21} \text{ m}^{-3}.$$

Then

$$\ell_{pm} = \frac{1}{(5.6 \times 10^{21})(10^{-20})} \simeq 1.8 \times 10^{-2} \text{ m}.$$

Final answer.

$$\ell_{pm} \simeq 2 \times 10^{-2} \text{ m} \simeq 2 \text{ cm}.$$

Question 21 – Can a resting electron ionize nitrogen?

The electric field is $E \simeq 100 \text{ V m}^{-1}$. Over one mean free path, the potential difference is

$$\Delta V = E \ell_{pm} \simeq 100 \times 0.018 \simeq 1.8 \text{ V}.$$

An electron crossing a potential difference of 1.8 V gains 1.8 eV. The ionization energy of nitrogen is

$$\mathcal{E}_i \simeq 16 \text{ eV}.$$

So the typical energy gained before a collision is far too small.

The distance needed to gain 16 eV is

$$x_i = \frac{16 \text{ V}}{100 \text{ V m}^{-1}} = 0.16 \text{ m}.$$

In units of the mean free path,

$$\frac{x_i}{\ell_{pm}} \simeq \frac{0.16}{0.018} \simeq 9.$$

The probability of travelling that far without collision is roughly $e^{-9} \sim 10^{-4}$. Such a long free path is possible, but rare.

Final answer. A resting electron typically gains only about 2 eV before collision, much less than 16 eV. It will not normally ionize N_2 ; only rare paths of about nine mean free paths, with probability $e^{-9} \sim 10^{-4}$, could do it.

Question 22 – Why 1 MeV cosmic electrons are relativistic

The rest energy of an electron is

$$m_e c^2 \simeq 0.511 \text{ MeV}.$$

A kinetic energy of 1 MeV is about twice the rest energy, so the non-relativistic approximation cannot be valid. The cleanest way to show that it is *necessarily* invalid is to assume the opposite and reach a contradiction: with $K = \frac{1}{2}m_e v^2$,

$$v = \sqrt{\frac{2K}{m_e}} = \sqrt{\frac{2 \times 1 \times 10^6 \times 1.60 \times 10^{-19}}{9.11 \times 10^{-31}}} \simeq 5.9 \times 10^8 \text{ m s}^{-1} \simeq 2c,$$

which is impossible. A 1 MeV electron must therefore be treated relativistically.

The Lorentz factor is

$$\Gamma = 1 + \frac{K}{m_e c^2} \simeq 1 + \frac{1}{0.511} \simeq 2.96 \simeq 3.$$

Then

$$\frac{v}{c} = \sqrt{1 - \frac{1}{\Gamma^2}} \simeq 0.94.$$

Final answer. A 1 MeV electron has $\Gamma \simeq 3$ and $v \simeq 0.94c$, so it is necessarily relativistic.

Question 23 – Cascade avalanche explanation

A high-energy primary electron can ionize a nitrogen molecule during a collision. That collision releases a secondary electron. After the first ionization, there are more free electrons than before. These electrons can themselves be accelerated and produce more ionizations.

Fix the counting convention that Question 24 will use: one primary electron is present at stage 0, and an impact ionization leaves the incident electron free while releasing one new electron from the molecule, so an ideal stage doubles the population:

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow \dots, \quad N_e(\mathcal{N}) \simeq 2^{\mathcal{N}}.$$

This is exponential growth. The energy is supplied by the electric field, which re-accelerates every electron between two collisions, and by the energetic primary particle. The process is called an avalanche because each generation creates more particles able to create the next generation.

The avalanche cannot grow forever. It can be limited by energy losses, recombination, attachment to neutral molecules, or screening of the electric field by the newly produced charges. During the process, excited molecules can emit visible light, which contributes to the sprite glow.

Final answer. A cascade avalanche is a chain reaction of impact ionizations. One energetic electron creates secondary electrons, which create more ionizations, leading to approximately exponential growth of electron number and current.

Question 24 – Avalanche stage number

We want the average electron density to reach

$$n = 1 \times 10^{11} \text{ m}^{-3}.$$

Let $\mathcal{N}$ be the number of avalanche stages. In a doubling model, the number of electrons after $\mathcal{N}$ stages is approximately

$$2^{\mathcal{N}}.$$

The problem states that the electrons are contained in a characteristic volume

$$(\mathcal{N}\ell_{pm})^3.$$

Using $\ell_{pm} \simeq 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$ from Question 20,

$$n \simeq \frac{2^{\mathcal{N}}}{(\mathcal{N}\ell_{pm})^3}.$$

Set this equal to $1 \times 10^{11} \text{ m}^{-3}$:

$$10^{11} = \frac{2^{\mathcal{N}}}{\mathcal{N}^3(2 \times 10^{-2})^3}.$$

So

$$2^{\mathcal{N}} = 10^{11} \mathcal{N}^3 \times 8 \times 10^{-6} = 8 \times 10^5 \mathcal{N}^3.$$

Equivalently,

$$2^{\mathcal{N}-3} = 10^5 \mathcal{N}^3.$$

Multiplying by 10^{-9} gives

$$10^{-9} \times 2^{\mathcal{N}-3} = 10^{-4} \mathcal{N}^3.$$

This is exactly the equality represented by the two curves in Figure 7.

It is worth being explicit about the value of ℓ_{pm} used here, because the statement says “of the order of one centimetre” while the curves of Figure 7 are calibrated for the 2 cm obtained in Question 20. Running the algebra backwards from the plotted functions,

$$10^{-9} 2^{\mathcal{N}-3} = 10^{-4} \mathcal{N}^3 \iff 2^{\mathcal{N}} = 8 \times 10^5 \mathcal{N}^3 \iff n = \frac{2^{\mathcal{N}}}{\mathcal{N}^3 \ell_{pm}^3} = 10^{11} \text{ with } \ell_{pm} = 2 \text{ cm},$$

since $10^{11} \ell_{pm}^3 = 10^{11} \times 8 \times 10^{-6} = 8 \times 10^5$. Taking ℓ_{pm} to be exactly 1 cm would instead give $2^{\mathcal{N}} = 10^5 \mathcal{N}^3$ and $\mathcal{N} \simeq 32$, and the two plotted curves could not be used at all. The statement’s “of the order of one centimetre” should therefore be read as the order of magnitude of the 1.8 cm found in Question 20, and the graph should be used with that value. Either reading gives a few tens of stages, so the conclusion is robust.

Procedure on the graph: plot the two functions on the same axes, read the abscissa of their intersection. The exponential f_1 is negligible until $\mathcal{N} \simeq 28$, then crosses the slowly growing cubic f_2 a little beyond $\mathcal{N} = 32$, at an ordinate close to 4.3.

At $\mathcal{N} = 35$,

$$10^{-9} 2^{35-3} \simeq 4.29, \quad 10^{-4} 35^3 \simeq 4.29,$$

so the intersection is genuinely near

$$\mathcal{N} \simeq 35.$$

The corresponding length scale is

$$\mathcal{N}\ell_{pm} \simeq 35 \times 2 \text{ cm} \simeq 0.7 \text{ m}.$$

Final answer.

$$\mathcal{N} \simeq 35,$$

i.e. a few tens of stages, developing over a length $\mathcal{N}\ell_{pm} \simeq 0.7 \text{ m}$. (Note that 35 sits exactly on a rounding boundary, so quoting “ 4×10^1 ” to one significant figure is misleading here; $\mathcal{N} \simeq 35$ is the meaningful statement, and the literal 1 cm reading would give $\mathcal{N} \simeq 32$.)

Final checked results

Quantity	Checked value
Pic du Midi–Alps distance order	$d \sim 500 \text{ km}$
Curvature height over that distance	$h \sim 20 \text{ km}$ ($\sim 30 \text{ km}$ for Mont Blanc at 620 km)
Gravity at ISS altitude	$G_{\text{ISS}} \sim 9 \text{ m s}^{-2}$
ISS speed	$v_{\text{ISS}} \sim 8 \text{ km s}^{-1}$
ISS period	$T_{\text{ISS}} \sim 90 \text{ min}$
Atmospheric scale height	$H = 8.5 \text{ km} \sim 8 \text{ km}$
Plasma cutoff for $n = 10^{11} \text{ m}^{-3}$	$f_p \sim 3 \text{ MHz}$
Radar cutoff density at 2.2 MHz	$n_{\text{cut}} \sim 6 \times 10^{10} \text{ m}^{-3}$
Mean free path at 60 km	$\ell_{pm} \sim 2 \text{ cm}$
Energy gained over one mean free path	$eE\ell_{pm} \sim 2 \text{ eV}$
Avalanche stage number	$\mathcal{N} \simeq 35$ (a few tens)